

## 1. Introduction to Preliminaries

We introduce some basic structures and concepts from mathematics that are important to the study of Ordinary Differential Equation (ODE). While these concepts may not be necessarily part of the syllabus, they are briefly mentioned here and you should look into some advanced mathematics books from the library to get yourself familiarized with these concepts. Any advanced mathematics book on metric spaces or real analysis should contain more materials on these topics. I would urge you to write some proofs and try to understand the fundamental idea and motivation behind these concepts.

Informally speaking, in mathematics we specify sets and impose some axioms on the members of the set. We call these sets with their axioms *structures*. These axioms are NOT random, they are chosen carefully and if the structure is proven helpful, over time they are included in standard textbooks. It is important you understand what structures you need to study Ordinary Differential Equations (ODEs) along with various methods of solving ODEs. If you just study the techniques or methods of finding solutions of the ODEs you will be able to solve simple ODEs but you will not be able to solve the complicated ones where the solution may not exist in the appropriate interval and you may use one of the methods to calculate a solution but upon closer inspection you will find some contradiction which will prove your solution to be wrong.

I am providing here a brief INFORMAL motivation for why each structure is being introduced before defining them. We begin with the structure called *field*. Let us begin by setting aside all our mathematics knowledge beyond the sets of natural numbers ( $\mathbb{N}$ ), rational numbers ( $\mathbb{Q}$ ), real numbers ( $\mathbb{R}$ ), and complex numbers ( $\mathbb{C}$ ). You will see, before we can study anything we would need to define what we mean by a scalar and a set of scalars. We would need a list of such quantities and maybe more appropriately a large set of these quantities that possess some properties that we know a scalar might possess, properties like “the sum of two scalars is also a scalar”, “the product of two scalars is also a scalar”, etc. We can define a set containing these scalars and we can impose all the known properties of scalars as axioms on the elements of this set. We can then be sure if such a set with the axioms are well defined the elements of the set will obey the axioms. We will then have a set of scalars and be sure the scalars obey all properties that make them a scalar. This is the intention behind the structure called *field*. You can consider it as a set of scalars with the properties of scalars imposed on them. Let us now formally define a *field*.

**Definition (Field):** Consider a set  $\mathbb{F}$  with two binary operations  $+\mathbb{F} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$  and  $*\mathbb{F} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$  defined on  $\mathbb{F}$ . We will denote these binary operations using their infix notations i.e.  $+\mathbb{F}(a, b)$  will be written as  $a +\mathbb{F} b$  and  $*\mathbb{F}(a, b)$  will be written as  $a *\mathbb{F} b$ . The operation  $+\mathbb{F}$  is generally referred to as *addition* and  $\alpha +\mathbb{F} \beta$  as the *sum of  $\alpha$  and  $\beta$* . The operation  $*\mathbb{F}$  is generally referred to as *multiplication* and  $\alpha *\mathbb{F} \beta$  as the *product of  $\alpha$  and  $\beta$* . The set  $\mathbb{F}$  along with two such operations  $+\mathbb{F}$  and  $*\mathbb{F}$ , denoted by  $(\mathbb{F}, +\mathbb{F}, *\mathbb{F})$ , will be considered a *field* if they satisfy the following axioms:

- For every pair of elements  $\alpha, \beta \in \mathbb{F}$ , there is a corresponding unique element  $\alpha +_{\mathbb{F}} \beta \in \mathbb{F}$  i.e. the operation  $+_{\mathbb{F}}$  is closed in  $\mathbb{F}$ .
1.  $\alpha +_{\mathbb{F}} \beta = \beta +_{\mathbb{F}} \alpha$  for all  $\alpha, \beta \in \mathbb{F}$ .
  2.  $(\alpha +_{\mathbb{F}} \beta) +_{\mathbb{F}} \gamma = \alpha +_{\mathbb{F}} (\beta +_{\mathbb{F}} \gamma)$  for all  $\alpha, \beta, \gamma \in \mathbb{F}$ .
  3. There exists an element  $0_{\mathbb{F}} \in \mathbb{F}$  such that  $0_{\mathbb{F}} +_{\mathbb{F}} \alpha = \alpha$  for all  $\alpha \in \mathbb{F}$ . This element  $0_{\mathbb{F}}$  is called the additive identity of the field.
  4. For every  $\alpha \in \mathbb{F}$  there exists an element  $\beta \in \mathbb{F}$  such that  $\alpha +_{\mathbb{F}} \beta = 0_{\mathbb{F}}$ . The element  $\beta$  is called the additive inverse of the element  $\alpha$ .
- For every pair of elements  $\alpha, \beta \in \mathbb{F}$ , there is a corresponding unique element  $\alpha *_{\mathbb{F}} \beta \in \mathbb{F}$  i.e. the operation  $*_{\mathbb{F}}$  is closed in  $\mathbb{F}$ .
5.  $\alpha *_{\mathbb{F}} \beta = \beta *_{\mathbb{F}} \alpha$  for all  $\alpha, \beta \in \mathbb{F}$ .
  6.  $(\alpha *_{\mathbb{F}} \beta) *_{\mathbb{F}} \gamma = \alpha *_{\mathbb{F}} (\beta *_{\mathbb{F}} \gamma)$  for all  $\alpha, \beta, \gamma \in \mathbb{F}$ .
  7. There exists an element  $1_{\mathbb{F}} \in \mathbb{F}$  with  $1_{\mathbb{F}} \neq 0_{\mathbb{F}}$  such that  $1_{\mathbb{F}} *_{\mathbb{F}} \alpha = \alpha$  for all  $\alpha \in \mathbb{F}$ . The element  $1_{\mathbb{F}}$  is called the multiplicative identity of the field.
  8. For every  $\alpha \in \mathbb{F}$  with  $\alpha \neq 0_{\mathbb{F}}$  there exists an element  $\beta \in \mathbb{F}$  such that  $\alpha *_{\mathbb{F}} \beta = 1_{\mathbb{F}}$ . The element  $\beta$  is called the multiplicative inverse of the element  $\alpha$ .
  9. The operation  $*_{\mathbb{F}}$  is distributive over  $+_{\mathbb{F}}$   $\alpha *_{\mathbb{F}} (\beta +_{\mathbb{F}} \gamma) = (\alpha *_{\mathbb{F}} \beta) +_{\mathbb{F}} (\alpha *_{\mathbb{F}} \gamma)$  for all  $\alpha, \beta, \gamma \in \mathbb{F}$ .

If there is no ambiguity the field  $(\mathbb{F}, +_{\mathbb{F}}, *_{\mathbb{F}})$ , can be denoted just by the set  $\mathbb{F}$ .

Here are some examples of fields

1. The set of all rational numbers  $\mathbb{Q}$  with the usual addition and multiplication operations forms a field, denoted as  $(\mathbb{Q}, +, \times)$ .
2. The set of all real numbers  $\mathbb{R}$  with the usual addition and multiplication operations forms a field, denoted as  $(\mathbb{R}, +, \times)$ .
3. The set of all complex numbers  $\mathbb{C}$  with the usual complex addition and complex multiplication operations forms a field, denoted as  $(\mathbb{C}, +, \times)$ .

In our study of ODEs we will keep our focus on the field of real numbers,  $(\mathbb{R}, +, \times)$ , and occasionally the field of complex numbers,  $(\mathbb{C}, +, \times)$ . We will in most cases, if there is no ambiguity, refer to them with just the symbols for the sets, like  $\mathbb{R}$  and  $\mathbb{C}$ .

We now have a formal definition for the scalars and their sets. We have also seen (if you have done the proofs) that the real numbers, complex numbers, rational numbers which we knew were scalars intuitively can now be formally written as element of a field like  $(\mathbb{R}, +, \times)$ ,  $(\mathbb{C}, +, \times)$ , etc. We can now use this formal definition to study ODEs in a rigorous fashion without ambiguity. I would urge you to look into any *Linear Algebra* book from the library and study some more properties of fields and try to prove them. Let us now define something else, the formal definition for an *absolute value function* on a field. This is necessary as we still don't have any mechanism to compare two members of a field and the *absolute value function* will map the members of the field to the set  $\mathbb{R}$  where we have a mechanism of ordering members. There are other reasons too and you can look into them in any *abstract algebra* book. Let us now formally define the *absolute value function* on a field.

**Definition (Absolute Value Function):** Consider a field  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$  and a function  $|\cdot|_{\mathbb{K}} : \mathbb{K} \rightarrow \mathbb{R}$ . This function is called an *absolute value function* on the field  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$  if it satisfies the following conditions

1.  $|x|_{\mathbb{K}} \geq 0$  for all  $x \in \mathbb{K}$ , and  $|x|_{\mathbb{K}} = 0$  if and only if  $x = 0_{\mathbb{K}}$ .
2.  $|x *_{\mathbb{K}} y|_{\mathbb{K}} = |x|_{\mathbb{K}} |y|_{\mathbb{K}}$  for all  $x, y \in \mathbb{K}$
3.  $|x +_{\mathbb{K}} y|_{\mathbb{K}} \leq |x|_{\mathbb{K}} + |y|_{\mathbb{K}}$  for all  $x, y \in \mathbb{K}$

The value  $|x|_{\mathbb{K}}$  for any  $x \in \mathbb{K}$  is called the *absolute value* of  $x$ . If there is no ambiguity, we can just use  $|\cdot|$  to denote the function.

Here are a few examples of a few common absolute value functions

1. The function  $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$  defined on the field  $(\mathbb{R}, +, \times)$  as  $|x| := x$  if  $x \geq 0$  else  $|x| := -x$ .
2. The function  $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$  defined on the field  $(\mathbb{C}, +, \times)$  as  $|x| := \sqrt{x \times \bar{x}}$

We now have a formal definition of a field and a formal definition of the absolute value function and for our purposes we have enough tools to deal with scalars for the moment. We now need a way to formalize vectors.

I will provide a brief INFORMAL motivation for the formal definition of vectors but I urge you to go through any *Linear Algebra* book in the library for a more detailed exposition. In our current scenario we have a formal definition for a field and we know the basic sets like  $\mathbb{R}, \mathbb{N}, \mathbb{Q}, \mathbb{C}$ , etc. We can use the field structure to define and use a set of scalars but if you think about it there are no mechanism to represent and work with vectors. We can try and use the field structure for the vectors as well, but we will quickly realize we don't have a viable set of vectors for instance, if we use the field structure for formally defining a set of vectors those vectors will need a separate function for the concept of scalar multiplication and eventually a scaled vector. We therefore require a new mechanism for formally defining vectors. We stick with the structure approach where we take a set and a few axioms and impose these axioms on the elements of the set. These axioms may describe properties that we intuitively know vectors possess like "*The sum of two vectors is a vector*", "*The product of a scalar and a vector is just a scaled vector*", etc. If we choose the axioms properly a viable structure can be created which can be used to formally define a set of vectors. This is the intention behind the structure called a *vector space*. It describes a set of vectors and a list of axioms the vectors must obey. let us now formally define a *vector space*.

**Definition (Vector Space):** Consider a set  $\mathbb{V}$  and a field  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$  with two binary operations  $+_{\mathbb{V}} : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{V}$  and  $*_{\mathbb{V}} : \mathbb{K} \times \mathbb{V} \rightarrow \mathbb{V}$  defined on  $\mathbb{V}$ . We will denote these binary operations using their infix notations i.e.  $+_{\mathbb{V}}(a, b)$  will be written as  $a +_{\mathbb{V}} b$  and  $*_{\mathbb{V}}(a, b)$  will be written as  $a *_{\mathbb{V}} b$ . The set  $\mathbb{V}$  along with any two operations  $+_{\mathbb{V}}$  and  $*_{\mathbb{V}}$ , denoted by  $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}})$ , will be considered a *vector space over the field*  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$  if they satisfy the following axioms:

- For every pair of elements  $a, b \in \mathbb{V}$ , there is a corresponding unique element  $a +_{\mathbb{V}} b \in \mathbb{V}$  i.e. the operation  $+_{\mathbb{V}}$  is closed in  $\mathbb{V}$ .
  1.  $a +_{\mathbb{V}} b = b +_{\mathbb{V}} a$  for all  $a, b \in \mathbb{V}$ .
  2.  $(a +_{\mathbb{V}} b) +_{\mathbb{V}} c = a +_{\mathbb{V}} (b +_{\mathbb{V}} c)$  for all  $a, b, c \in \mathbb{V}$ .
  3. There exists an element  $0_{\mathbb{V}} \in \mathbb{V}$  such that  $0_{\mathbb{V}} +_{\mathbb{V}} a = a$  for all  $a \in \mathbb{V}$ . The element  $0_{\mathbb{V}}$  is called the *additive identity* of the vector space.
  4. For every  $a \in \mathbb{V}$  there exists an element  $b \in \mathbb{V}$  such that  $a +_{\mathbb{V}} b = 0_{\mathbb{V}}$ . The element  $b$  is called the *additive inverse* of the element  $a$ . It is often written as  $-a$ .
- For every  $\alpha \in \mathbb{K}$  and  $a \in \mathbb{V}$ , there is a corresponding unique element  $\alpha *_{\mathbb{V}} a \in \mathbb{V}$ .
  5.  $(\alpha *_{\mathbb{K}} \beta) *_{\mathbb{V}} a = \alpha *_{\mathbb{V}} (\beta *_{\mathbb{V}} a)$  for all  $\alpha, \beta \in \mathbb{K}$  and  $a \in \mathbb{V}$ .
  6.  $1_{\mathbb{K}} *_{\mathbb{V}} a = a$  for all  $a \in \mathbb{V}$  and where  $1_{\mathbb{K}}$  is the multiplicative identity of the field  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$ .
  7. The operation  $*_{\mathbb{V}}$  is distributive over  $+_{\mathbb{V}}$  i.e.,  $\alpha *_{\mathbb{V}} (a +_{\mathbb{V}} b) = (\alpha *_{\mathbb{V}} a) +_{\mathbb{V}} (\alpha *_{\mathbb{V}} b)$  for all  $\alpha \in \mathbb{K}$  and  $a, b \in \mathbb{V}$ .
  8. The operation  $*_{\mathbb{V}}$  is distributive over  $+_{\mathbb{K}}$  i.e.,  $(\alpha +_{\mathbb{K}} \beta) *_{\mathbb{V}} a = (\alpha *_{\mathbb{V}} a) +_{\mathbb{V}} (\beta *_{\mathbb{V}} a)$  for all  $\alpha, \beta \in \mathbb{K}$  and  $a \in \mathbb{V}$ .

If there is no ambiguity the vector space  $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}})$  over the field  $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$  can just be denoted by the set  $\mathbb{V}$ . The operation  $+_{\mathbb{V}}$  is generally referred to as *vector addition* and  $\alpha +_{\mathbb{V}} \beta$  as the *vector sum of  $\alpha$  and  $\beta$* . The operation  $*_{\mathbb{V}}$  is generally referred to as *scalar multiplication* and  $\alpha *_{\mathbb{V}} \beta$  as the *scalar product of  $\alpha$  and  $\beta$* .

Here are some examples of vector spaces

1. For any  $n \in \mathbb{N}$  with  $1 \leq n < \infty$ , the set  $\mathbb{R}^n$  or  $\mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$  ( $n$  times), the set of ordered  $n$ -tuples of real numbers forms a vector space over the field  $(\mathbb{R}, +, \times)$ . The vector addition operation here is defined as the usual addition operation used component-wise as  $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$  and the scalar multiplication is defined as the usual multiplication operation defined component-wise as  $\alpha \times (x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$  for all  $\alpha \in \mathbb{R}$  and  $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$ .
2. For any  $n \in \mathbb{N}$  with  $1 \leq n < \infty$ , the set  $\mathbb{C}^n$  or  $\mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$  ( $n$  times), the set of ordered  $n$ -tuples of complex numbers forms a vector space over the field  $(\mathbb{C}, +, \times)$ . The vector addition operation here is defined as the usual complex addition operation used component-wise as  $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$  and the scalar multiplication is defined as the usual complex multiplication operation defined component-wise as  $\alpha \times (x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$  for all  $\alpha \in \mathbb{C}$  and  $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{C}^n$ .
3. For any  $n \in \mathbb{N}$  with  $1 \leq n < \infty$ , the set  $\mathbb{C}^n$  or  $\mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$  ( $n$  times), the set of ordered  $n$ -tuples of complex numbers forms a vector space over the field  $(\mathbb{R}, +, \times)$ . The vector addition operation here is defined as the usual complex addition operation used component-wise as  $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$  and the scalar multiplication is defined as the usual multiplication operation defined component-wise as  $\alpha \times (x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$  for all  $\alpha \in \mathbb{R}$  and  $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{C}^n$ .
4. For  $a, b \in \mathbb{R}$  with  $a < b$ , consider the closed and bounded interval  $[a, b] \subset \mathbb{R}$ . The set of all continuous real-valued functions defined on the interval  $[a, b]$  (any function  $f : [a, b] \rightarrow \mathbb{R}$  which is continuous), denoted by  $\mathcal{C}([a, b])$ , forms a vector space over the field  $(\mathbb{R}, +, \times)$  with the vector addition operation defined as the usual addition operation defined pointwise on  $[a, b]$  as  $(f + g)(x) := f(x) + g(x)$  and the scalar multiplication defined as the usual multiplication operation also defined pointwise as  $(\alpha \times f)(x) := \alpha \times f(x)$  for all  $x \in [a, b]$ ,  $\alpha \in \mathbb{R}$ , and  $f, g \in \mathcal{C}([a, b])$ .

There are many such vector spaces and I would urge you to look into any advanced mathematics text book on *Linear Algebra* for more examples and more information on vector spaces.

We now have a formal definition of a vector but we do not have any mechanisms for obtaining the magnitude of the vector. This is an important tool as we can use it to compare vectors as well. If you think about it, you might guess that we need some form of an absolute value function for vectors. We do have something similar and it is called the *norm of a vector*. Let us formally define this norm function and also relabel a vector space which is equipped with such a function.

**Definition (Normed vector Space):** A vector space,  $\mathbb{V}$  over the field  $\mathbb{K}$ , is a *normed vector space*  $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}}; \|\cdot\|_{\mathbb{V}})$ , if there exists a function  $\|\cdot\|_{\mathbb{V}} : \mathbb{V} \rightarrow \mathbb{R}$  which satisfies the following conditions

1.  $\|x\|_{\mathbb{V}} \geq 0$  for all  $x \in \mathbb{V}$  and  $\|x\|_{\mathbb{V}} = 0$  if and only if  $x = 0_{\mathbb{V}}$ .
2.  $\|x +_{\mathbb{V}} y\|_{\mathbb{V}} \leq \|x\|_{\mathbb{V}} + \|y\|_{\mathbb{V}}$  for all  $x, y \in \mathbb{V}$ .
3.  $\|\alpha *_{\mathbb{V}} x\|_{\mathbb{V}} = \|\alpha\|_{\mathbb{K}} \|x\|_{\mathbb{V}}$  for all  $\alpha \in \mathbb{K}$ ,  $x \in \mathbb{V}$ .

We call  $\|\cdot\|_{\mathbb{V}}$  the *norm* of the vector space  $\mathbb{V}$ .

We now have a formal definition of a field, vector space, and a norm. These structures will be used regularly in the study of ODEs. However, you might notice that all the structures we have defined so far include a lot of algebraic properties in their axioms. You might wonder if a structure can be

constructed where the algebraic properties are not included but the fundamental idea of a set with a mechanism of comparing its elements is present. The answer would be yes there are structures that formalize this idea. Let us now formally define such a structure.

**Definition (Metric Space):** Let us consider a set  $\mathcal{M}$  with  $\mathcal{M} \neq \emptyset$  and a function  $d : \mathcal{M} \times \mathcal{M} \rightarrow \mathbb{R}$ , which satisfies the following conditions:

1.  $d(x, y) = d(y, x)$  for all  $x, y \in \mathcal{M}$ .
2.  $d(x, y) \geq 0$  for all  $x, y \in \mathcal{M}$  and  $d(x, y) = 0$  if and only if  $x = y$ .
3.  $d(x, y) \leq d(x, z) + d(z, y)$  for all  $x, y, z \in \mathcal{M}$ .

The function  $d$  is then called a *metric* and the set  $(\mathcal{M}, d)$  is called a *metric space*. If there is no ambiguity we generally denote the metric space with just the set  $\mathcal{M}$ .

We have now formally introduced a new structure called a *metric space*. We will need this structure for studying ODEs as this structure provides the basic fundamental platform for concepts like *convergence*, *limits*, *completeness*, etc. These concepts are of utmost importance when studying ODEs. Now to understand how metric spaces work with vector spaces (we only explore this upto our syllabus requirements, you may want to look up *Topological Vector Spaces* to learn more) I would urge you to do the following exercise yourself. You can use any *Linear Algebra* or *Real Analysis* book for solving the problem.

**Exercise:** Show that any normed vector space, say  $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}}; \|\cdot\|_{\mathbb{V}})$ , can also be considered a metric space with the metric defined as  $d(x, y) := \|x - y\|_{\mathbb{V}}$ .

We now define a list of properties that we will need when studying ODEs. I have not provided the motivation for these concepts yet but I would urge you to try to understand them and see if you can figure out the motivation. I will ofcourse revisit them and provide their motivation later when we encounter existence and uniqueness proofs.

**Definition (Convergence):** A sequence  $(x_i)_{i=1}^{\infty}$  in a metric space  $(\mathcal{M}, d)$ , is said to converge to an element  $x \in \mathcal{M}$  if  $\lim_{i \rightarrow \infty} d(x_i, x) = 0$ . We shall denote the point  $x$  as  $\lim_{i \rightarrow \infty} x_i$ .

**Definition (Cauchy Sequence):** A sequence  $(x_n)_{n=1}^{\infty}$  in a metric space  $(\mathcal{M}, d)$ , is called a *Cauchy sequence* if for any  $\varepsilon \in \mathbb{R}$  with  $\varepsilon > 0$ , there exists an  $N_{\varepsilon} \in \mathbb{N}$  with  $N_{\varepsilon} > 0$  such that  $d(x_m, x_n) < \varepsilon$  for all  $n, m \in \mathbb{N}$  with  $n, m \geq N_{\varepsilon}$ .

**Definition ( $\varepsilon$ -neighborhood):** Let us consider a metric space  $(\mathcal{M}, d)$ , an element  $x \in \mathcal{M}$ , and  $\varepsilon \in \mathbb{R}$  with  $\varepsilon > 0$ . The set  $\{z \in \mathcal{M} : d(x, z) < \varepsilon\}$  is defined as the  $\varepsilon$ -neighborhood of the point  $x$ .

**Definition (open set):** Let us consider a metric space  $(\mathcal{M}, d)$  and a set  $S \subseteq \mathcal{M}$ . If for all  $x \in S$  there exists an  $\varepsilon \in \mathbb{R}$  with  $\varepsilon > 0$  such that the  $\varepsilon$ -neighborhood of  $x$  is a subset of  $S$ , then  $S$  is called an *open set* and in this particular case an *open subset* of  $\mathcal{M}$ .

**Definition (Complete Metric Space):** A metric space  $(\mathcal{M}, d)$  is called a *complete metric space* if all cauchy sequences in  $\mathcal{M}$  converge to a point in  $\mathcal{M}$ .

**Definition (Banach Space):** A *Banach Space* is defined as a complete normed vector space.